

Multi-objectives and Surrogate-Based Optimisation

ENEL445 Applied Engineering Optimisation

2025

Outline

- Multi-Objective Optimisation
- Pareto Optimality
- Surrogate-Based Optimisation
- Maximum Likelihood

Outline

- **Multi-Objective Optimisation**
- Pareto Optimality
- Surrogate-Based Optimisation
- Maximum Likelihood



Speak UP
**STAND TOGETHER
STOP BULLYING**

KŌRERO MAI, KŌRERO ATU, MAURI TŪ, MAURI ORA



Multi-objective Optimisation

- Multi-objective optimisation problems - Problems whose formulations have more than one objective function:
 - Risk versus reward
 - Profit versus environmental impact
 - Acquisition cost versus operating cost
 - Drag versus noise
- A common technique when presented with multiple objectives is to assign weights to the various objectives and combine them into a single objective.
- More generally, multi-objective optimisation helps explore the trade-offs between different metrics.

Multi-objective Optimisation

- Example: In designing an aircraft, we may decide that maximising fuel tank volume and minimising weight are important. However, these metrics compete with each other and cannot be optimised simultaneously.
- Instead, we may conclude that maximising range (the distance the aircraft can fly) is the underlying metric that matters most for our application and appropriately balances the trade-offs between weight and fuel.
- Or perhaps maximising range is not the right metric. Range may be important, but only insofar as we reach some threshold. Increasing the range does not increase the value because the range is a constraint. The underlying objective in this scenario may be some other metric, such as operating costs.

Multi-objective Optimisation

- Multiobjective optimisation can quantify the trade-off between different objectives.
- Multiobjective optimisation provides a “family” of designs rather than a single design.
- For some problems, the underlying objective is either unknown or too difficult to compute.

Multi-objective Optimisation

- The objective function in a multi-objective optimisation can be expressed as

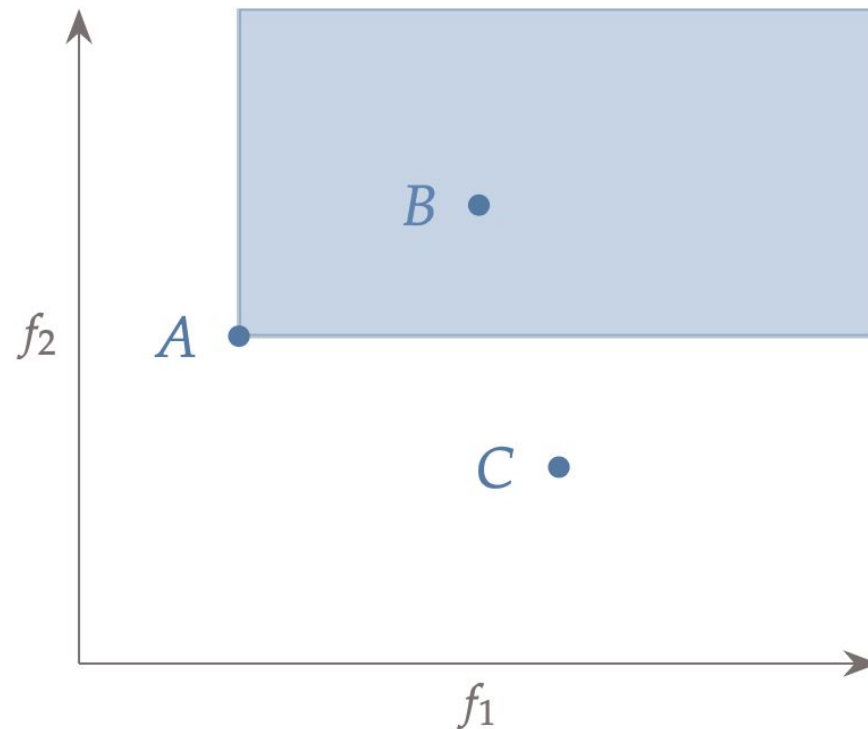
$$\underset{x}{\text{minimize}} \quad f(x) = \begin{bmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_{n_f}(x) \end{bmatrix}, \quad \text{where } n_f \geq 2.$$

- This multi-objective formulation may require trade-offs because, beyond some point, further reduction in one objective can only be achieved by increasing one or more of the other objectives.
- With multiple objectives, we have to reconsider what it means for a point to be optimal.

Outline

- Multi-Objective Optimisation
- **Pareto Optimality and Finding the Pareto Front**
- Surrogate-Based Optimisation
- Maximum Likelihood

Pareto Optimal



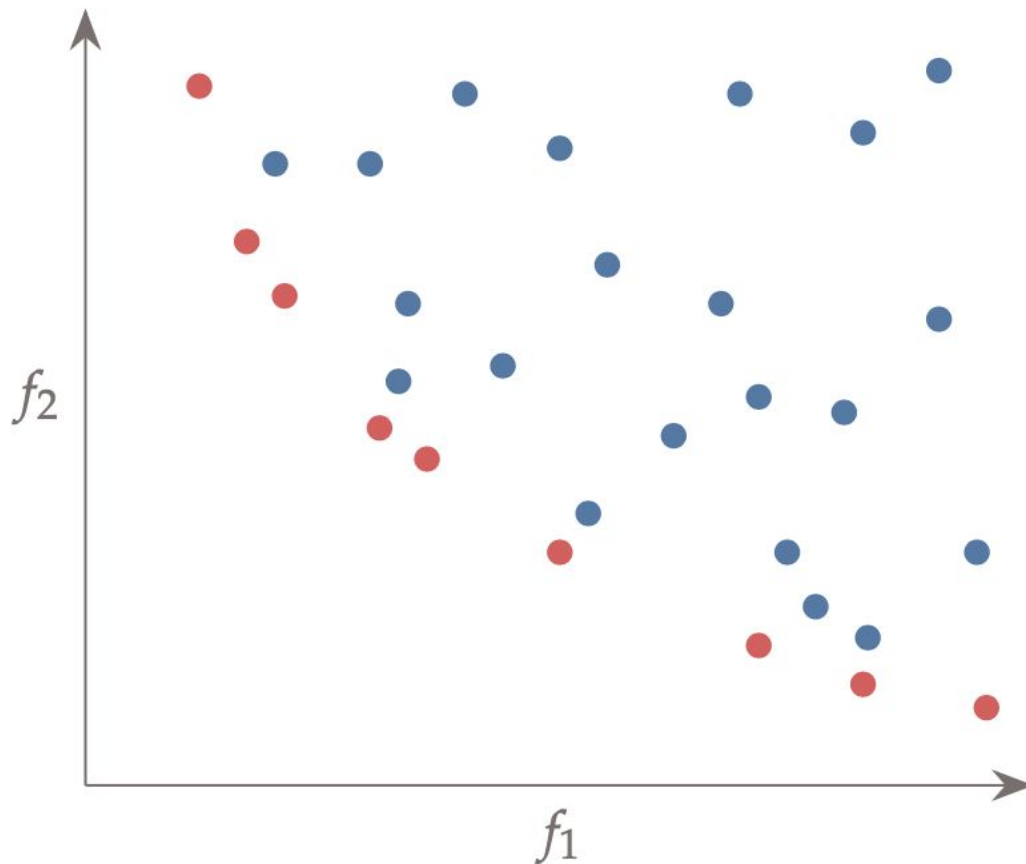
- One design is said to **dominate** another design if it is superior in all objectives (design A dominates any design in the shaded rectangle).
- A point is said to be **nondominated** if none of the other evaluated points dominate it

Pareto Optimal

- If a point is nondominated by any point in the entire domain, then that point is called **Pareto optimal**.
- This does not imply that this point dominates all other points; it simply means no other point dominates it.
- The set of all Pareto optimal points is called the **Pareto set**.

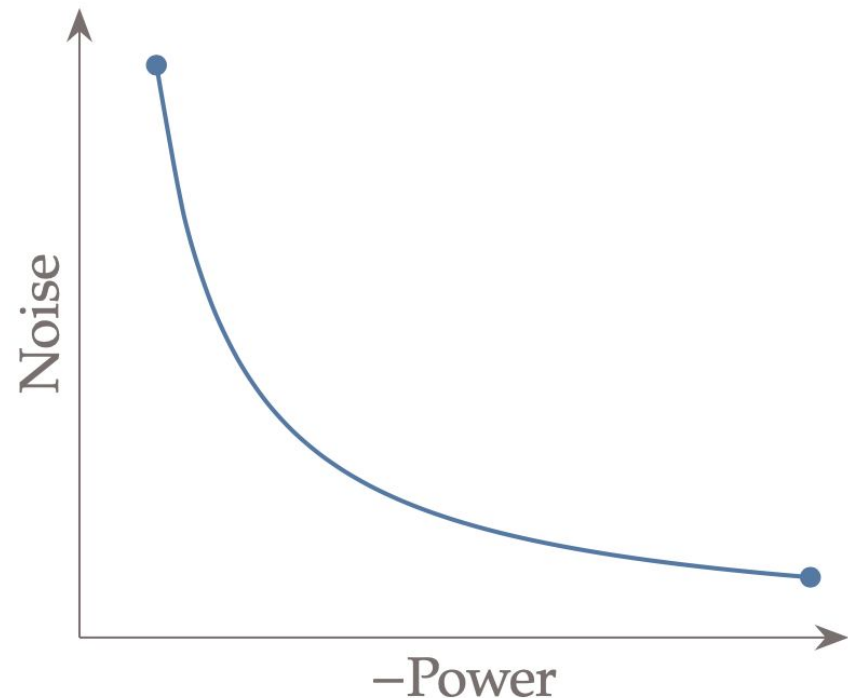
Pareto Front

- The Pareto set refers to the vector of points x^* , whereas the **Pareto front** refers to the vector of functions $f(x^*)$.



Example - Wind Farm Optimisation

- The two objectives are maximising power production and minimising noise.
- The Pareto front tells us how much power we have to sacrifice for a given reduction in noise. If the slope is steep, as is the case in the figure below, we can see that a small sacrifice in power production can be exchanged for significantly reduced noise.
- Understanding the magnitude of these trade-off sensitivities helps make high-level design decisions.



Finding the Pareto Front

- Various solution methods exist for solving multiobjective optimisation problems:
 - Weighted-sum method
 - Epsilon-constraint method
 - Normal Boundary Intersection (NBI) method
 - Evolutionary algorithms

Weighted Sum Method

- Combine all of the objectives into a single objective using a weighted sum:

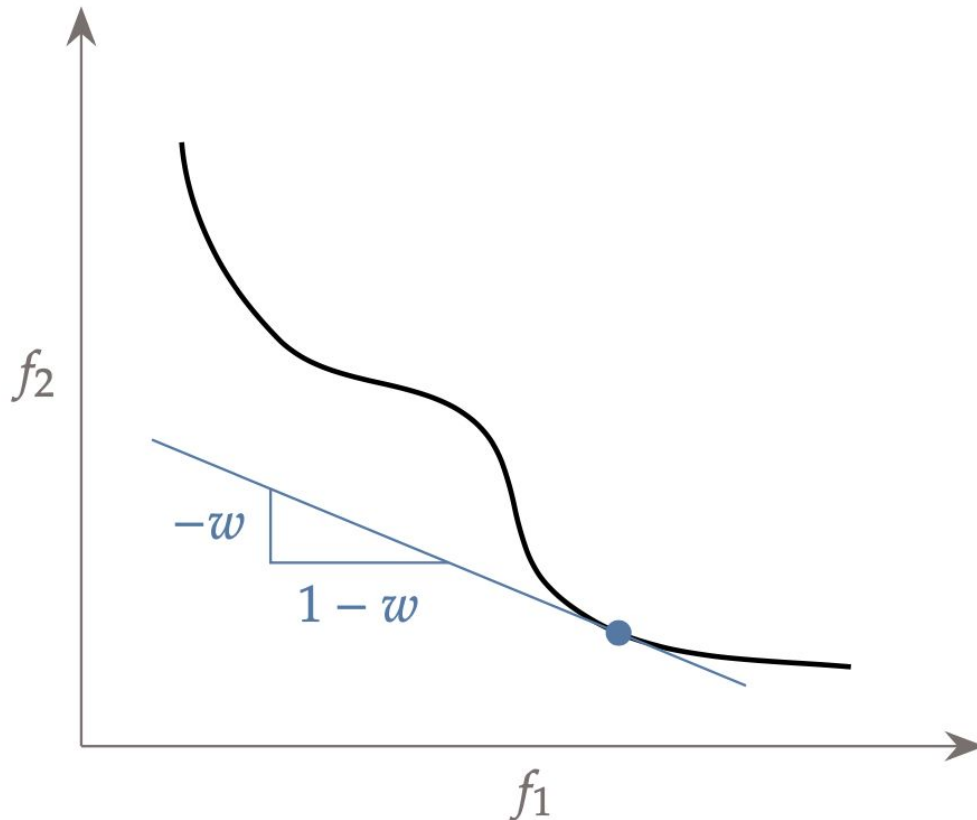
$$\bar{f}(x) = \sum_{i=1}^{n_f} w_i f_i(x)$$

where n_f is the number of objectives, and the weights are usually normalised such that

$$\sum_{i=1}^{n_f} w_i = 1$$

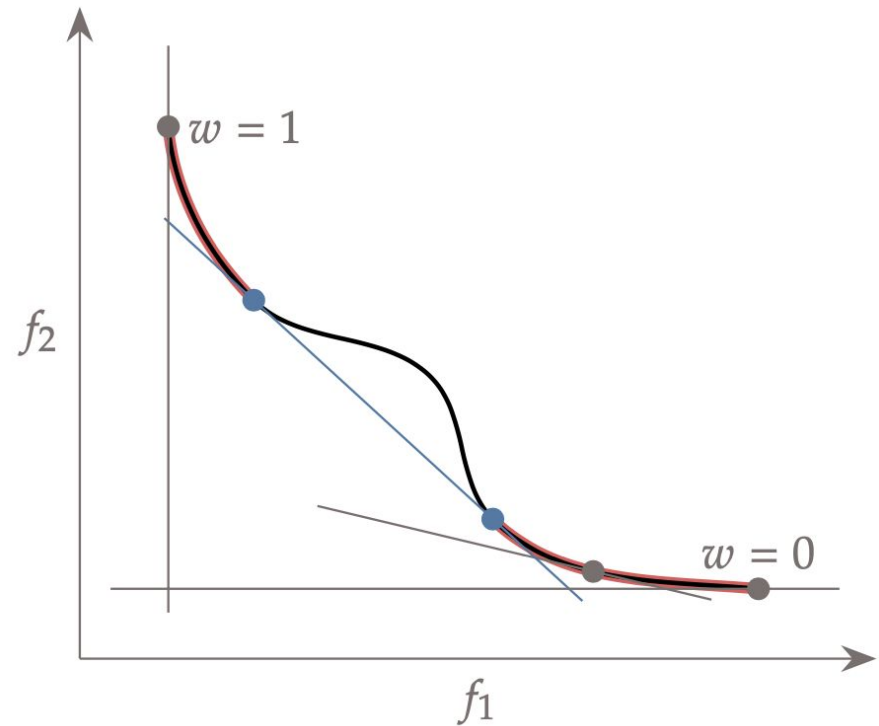
Weighted Sum Method

- Two objectives example:
- At the extremes $w = 0$ and $w = 1$, the optimisation returns the designs that optimise one objective while ignoring the other.



Weighted Sum Method

- Advantage:
 - It is easy to use
- Drawbacks:
 - Uniform spacing in w leads to non-uniform spacing along the Pareto set.
 - Not apparent which values of w should be used to sweep out the Pareto set evenly
 - This method can only return points on the convex portion of the Pareto front.



Epsilon-Constraint Method

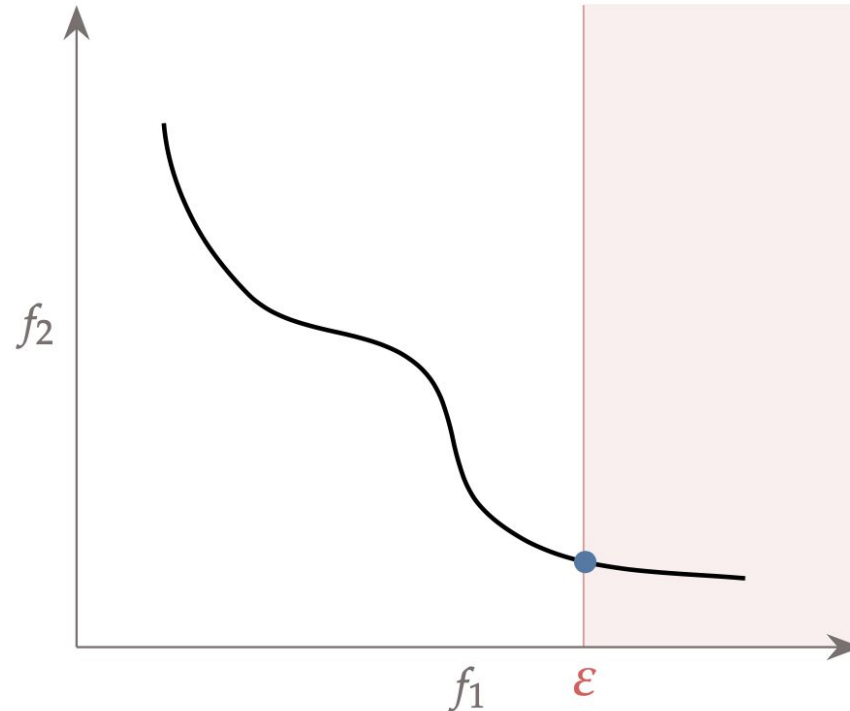
- Choose an ε for each constraint.
- Minimise one objective while setting all other objectives as additional constraints.

$$\begin{array}{ll}\text{minimize} & f_i \\ \text{subject to} & f_j \leq \varepsilon_j \quad \text{for all } j \neq i \\ & g(x) \leq 0 \\ & h(x) = 0.\end{array}$$

- Repeat this procedure for different values of ε_j

Epsilon-Constraint Method

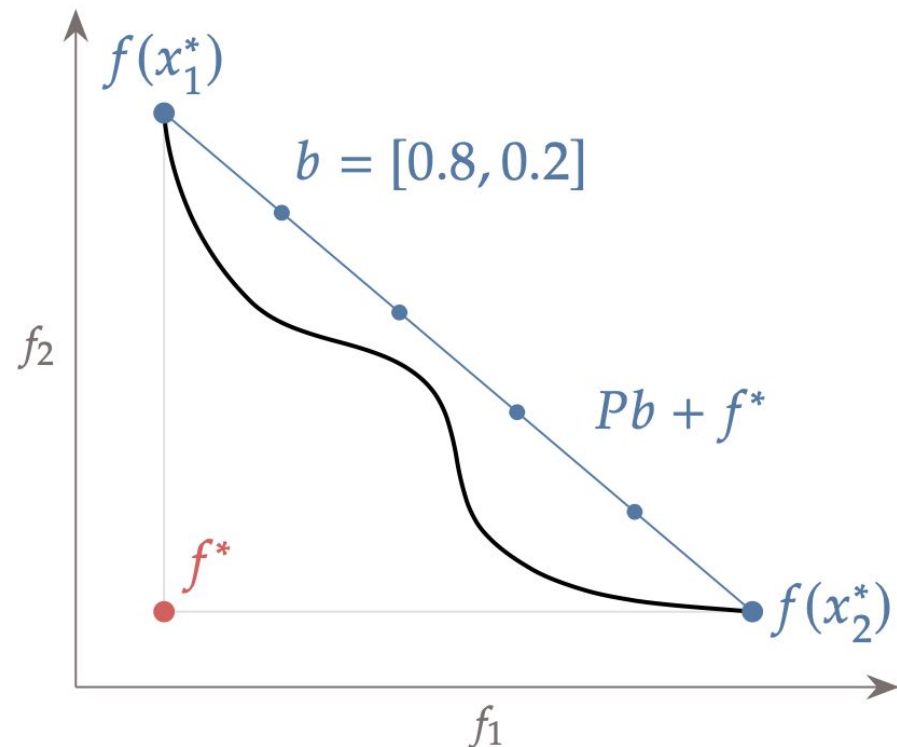
- Reveals the non-convex portions of the Pareto front.



- Like the weighted-sum method, a uniform spacing in ϵ does not generally yield a uniform spacing of the Pareto front (though it is usually much better spaced than weighted-sum) therefore it might still be inefficient, particularly with more than two objectives.

Normal Boundary Intersection

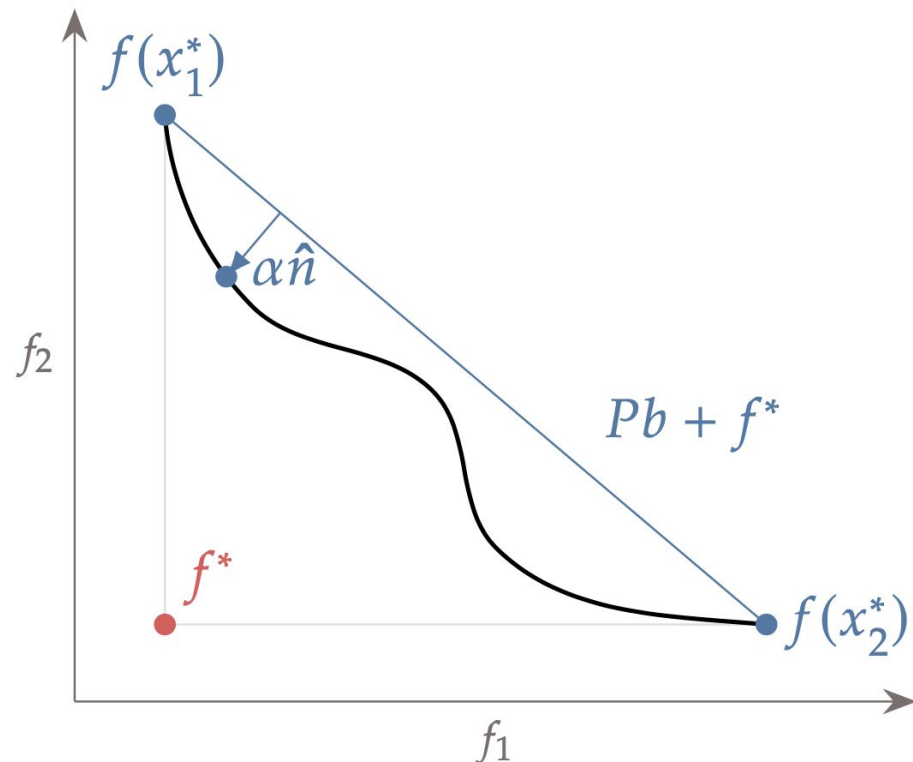
- The NBI method is designed to address the issue of nonuniform spacing along the Pareto front.
- First find the extremes of the Pareto set; in other words, minimise the objectives one at a time.
- Construct a plane that passes through the extreme points.
- Space points along this plane.
- Starting from those points, solve optimisation problems that search along directions normal to this plane.



Normal Boundary Intersection

- Mathematically, we start by determining the **anchor points**, which are just single-objective optimisation problems.
- From the anchor points, we define the **utopia point** - every objective reaches its minimum simultaneously.

$$f^* = \begin{bmatrix} f_1(x_1^*) \\ f_2(x_2^*) \\ \vdots \\ f_n(x_n^*) \end{bmatrix}$$



Normal Boundary Intersection

- We search along a direction perpendicular to the line defined by the anchor points. We seek to find the point along this direction that is the farthest away from the anchor points line (a maximisation problem), with the constraint that the point is consistent with the objective functions:

$$\begin{array}{ll}\text{maximize} & \alpha \\ & x, \alpha \\ \text{subject to} & Pb + f^* + \alpha \hat{n} = f(x) \\ & g(x) \leq 0 \\ & h(x) = 0.\end{array}$$

- The resulting optimal point found along this direction is a point on the Pareto front.
- We then repeat this process for another set of weighting parameters in b .

Normal Boundary Intersection

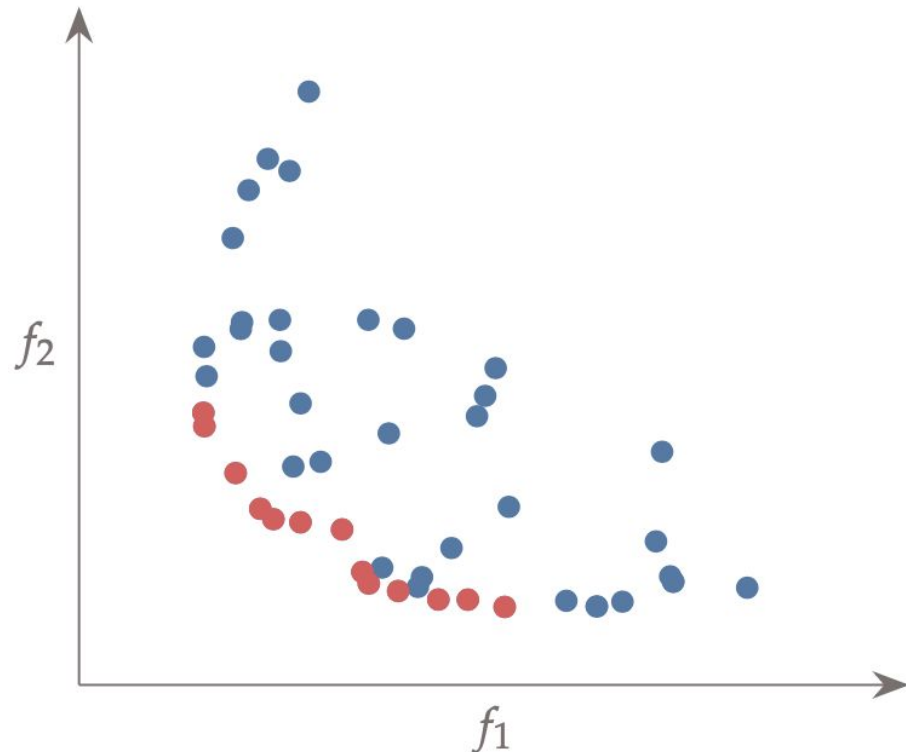
- The idea is that even spacing along this plane is more likely to lead to even spacing along the Pareto front.
- Yields a more uniformly spaced Pareto front
- Desirable for computational efficiency, albeit at the cost of a more complex methodology.
- For most multiobjective design problems, additional complexity beyond the NBI method is unnecessary. However, even this method can still have deficiencies for problems with unusual Pareto fronts, and new methods continue to be developed [1-3]. A more recent improvement performs an even more efficient generation of the Pareto front by avoiding regions of the Pareto front where minimal trade-offs occur [3].

Normal Boundary Intersection

- [1] Ismail-Yahaya and Messac, “Effective generation of the Pareto frontier using the normal constraint method”, 2002.
- [2] Messac and Mattson, “Normal constraint method with guarantee of even representation of complete Pareto frontier”, 2004.
- [3] Hancock and Mattson, “The smart normal constraint method for directly generating a smart Pareto set”, 2013.

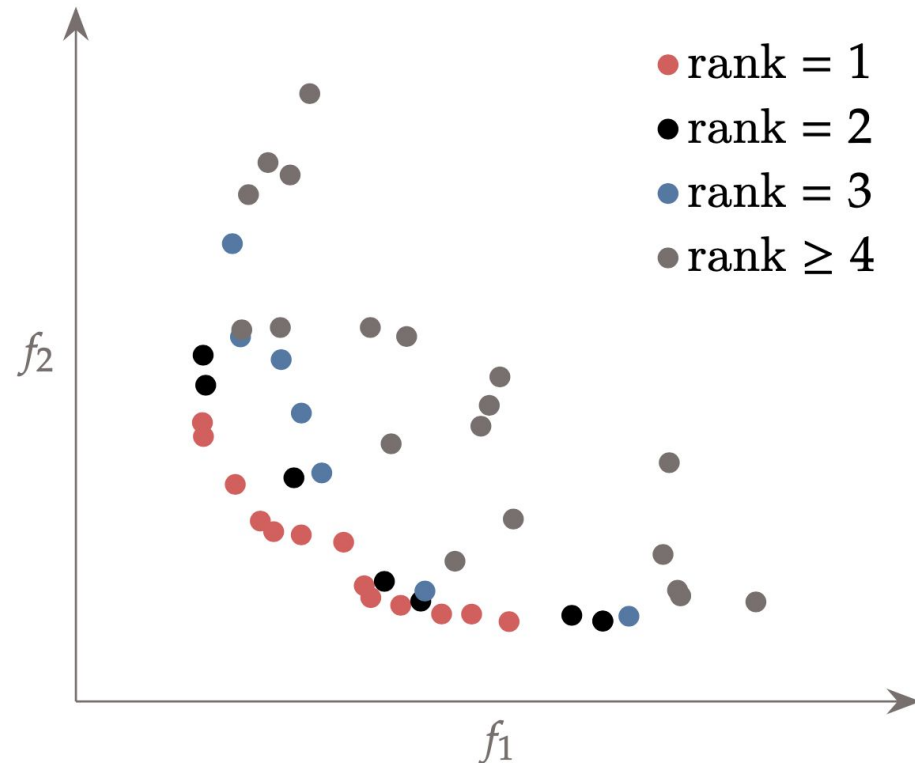
Evolutionary Algorithms

- A GA is amenable to an extension that can handle multiple objectives because it keeps track of a large population of designs at each iteration.
- If we plot two objective functions for a given population of a GA iteration, we get something shown in the figure below.
- The points represent the current population, and the highlighted points in the lower left are the current nondominated set. As the optimisation progresses, the nondominated set converges toward the actual Pareto front.



Evolutionary Algorithms

- Elitist Nondominated Sorting Genetic Algorithm (NSGA-II)
- In addition to determining the nondominated set, we want to rank all members by their **dominance depth**, which is also called **nondominated sorting**. In this approach, all nondominated points in the population (i.e., the current approximation of the Pareto set) are given a rank of 1. Those points are then removed from the set, and the next set of nondominated points is given a rank of 2, and so on.



Evolutionary Algorithms

- The main advantage of this multiobjective approach is that if an evolutionary algorithm is appropriate for solving a given single-objective problem, then the extra information needed for a multiobjective problem is already there, and solving the multiobjective problem does not incur much additional computational cost.
- However, solving multiple gradient-based problems may be more efficient than solving one gradient-free problem, especially for problems with a large number of design variables.

Multi-objective Optimisation - Summary

- Multiobjective optimisation is particularly useful in quantifying trade-off sensitivities between critical metrics. It is also useful when we seek a family of potential solutions rather than a single solution.
- Some scenarios where a family of solutions might be preferred include when the models used in optimisation are low fidelity and higher-fidelity design tools will be applied or when more investigation is needed.
- A multiobjective approach can produce candidate solutions for later refinement.

Outline

- Multi-Objective Optimisation
- Pareto Optimality and Finding the Pareto Front
- **Surrogate-Based Optimisation**
- Maximum Likelihood

Surrogate Models

- A **surrogate** is an approximate model that represents a curve fit to some underlying data. Also known as:
 - Response surface
 - Metamodel
- The goal of a surrogate model is to build a model that is much faster to compute than the original function, but still retains sufficient accuracy away from known data points.
- **Surrogate-based optimisation** (SBO) performs optimisation using the surrogate model.
- Instead of aiming for a globally accurate surrogate, SBO just needs the surrogate model to be accurate enough to lead the optimiser to the true optimum.

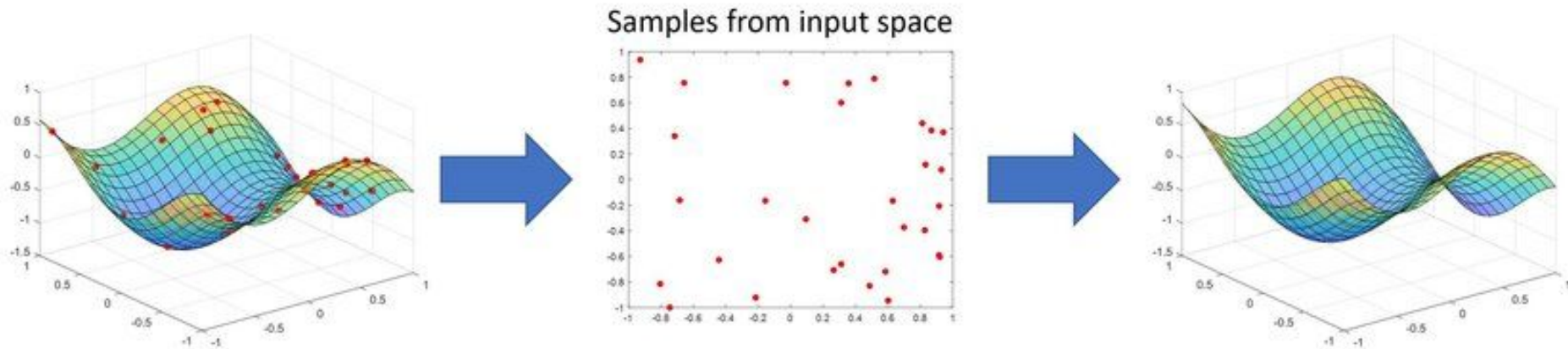


Figure from “Surrogate modelling for the forecast of Seveso-type atmospheric pollutant dispersion”

https://www.researchgate.net/publication/360665374_Surrogate_modelling_for_the_forecast_of_Seveso-type_atmospheric_pollutant_dispersion?tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6Ii9kaXJlY3QiLCJwYWdlIjoieX2RpcmVjdCJ9fQ

Why Surrogates

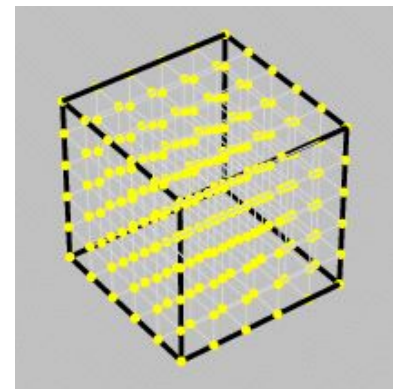
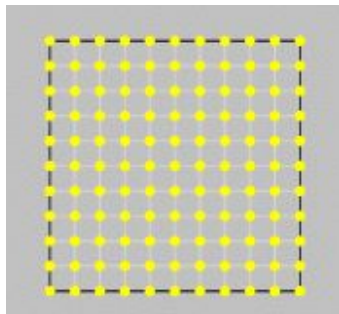
- This approach is often used when the function is expensive and the number of evaluations needed to build a sufficiently accurate surrogate model is less than that needed to optimise the original model directly.
- Surrogate modeling can be effective in handling noisy models because they create a smooth representation of noisy data.
- One scenario that leads to both expensive evaluation and noisy output is *experimental data*.
- Surrogate models are also helpful when we want to understand the design space. By constructing a continuous model over discrete data, we obtain functional relationships that can be visualised more effectively.

Why Surrogates

- A lower-fidelity surrogate can guide the optimiser to promising regions of the larger design space.
- Used in conjunction with other algorithms like GPS or Nelder-Mead.
- When multiple sources of data are available, surrogate models can fuse the data to build a single model. The data could come from numerical models with different levels of fidelity or experimental data. For example, surrogate models can calibrate numerical model data using experimental data. This is helpful because experimental data is usually much more scarce than numerical data. The same reasoning applies to low- vs high-fidelity numerical data.

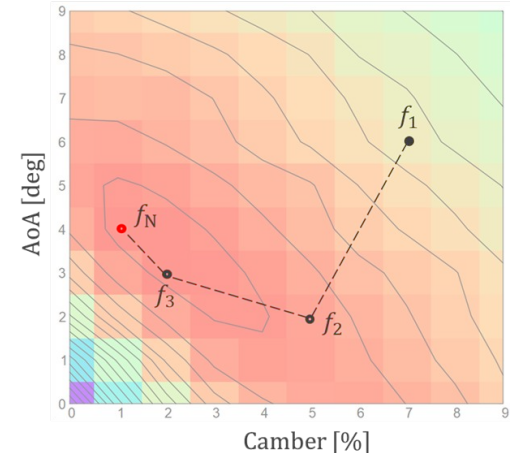
Downsides

- One potential issue with surrogate models is the **curse of dimensionality**, which refers to poor scalability with the number of design variables. The larger the number of design variables, the more model evaluations are needed to construct a surrogate model that is accurate enough. Therefore, the reasons for using surrogate models cited earlier might not be enough if the optimisation problem has a large number of design variables.



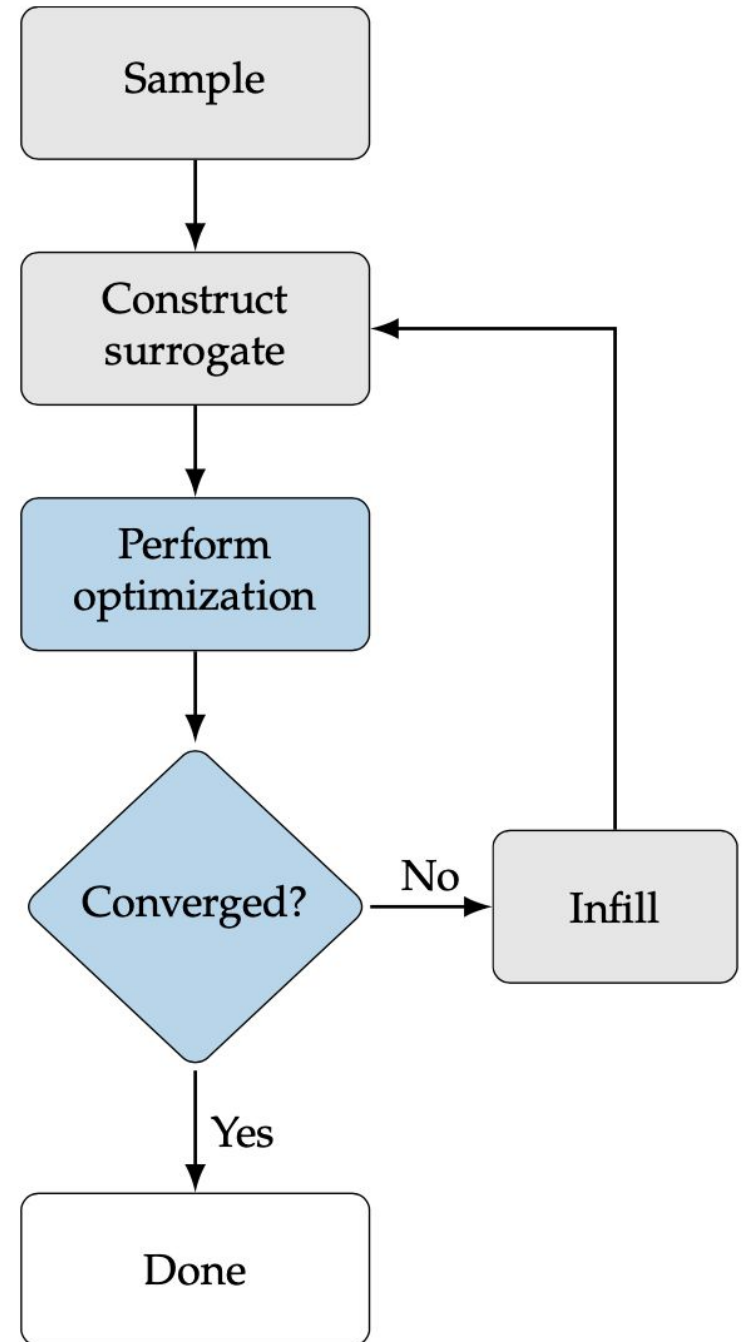
Surrogate Modelling Example

- When performing trajectory optimisation of an aircraft, we need to evaluate the lift and drag of the aircraft at each point of the trajectory.
- This typically requires many points, and computing the lift and drag at each point might be prohibitive. Therefore, it might be worthwhile to use surrogate models that predict the lift and drag as functions of the angle of attack.
- If the optimisation design variables include parameters that change the lift and drag characteristics, such as the wing shape, then the surrogate model needs to be rebuilt at every optimisation iteration.

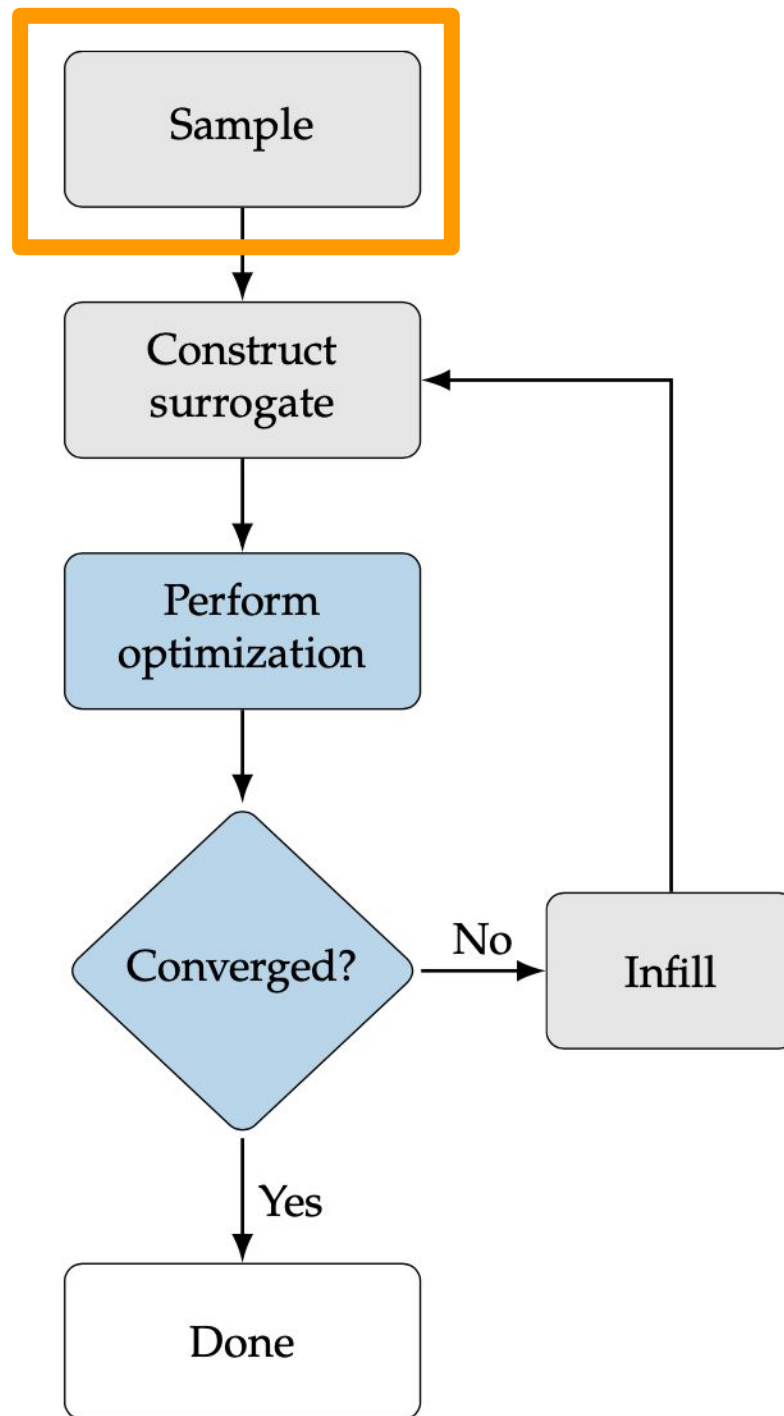


Surrogate Process

- Choose initial points to evaluate the function or conduct experiments. These points are sometimes referred to as **training data**. Next, we build a surrogate model from the sampled points. We can then perform optimisation by querying the surrogate model.
- Based on the results of the optimisation, we include additional points in the sample and reconstruct the surrogate (infill). We repeat this process until some convergence criterion or a maximum number of iterations is reached.

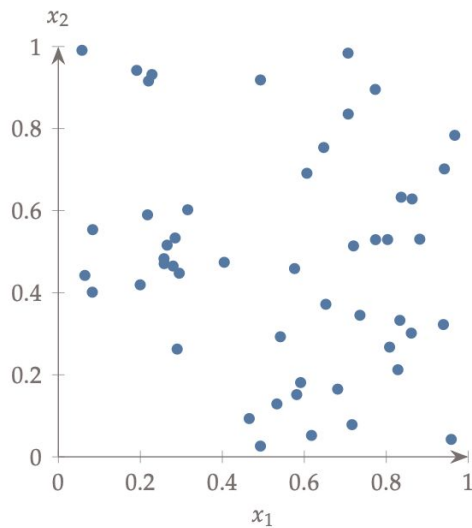


Surrogate Process

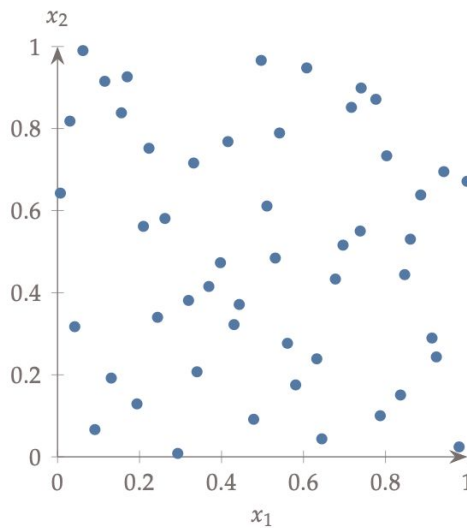


Sampling

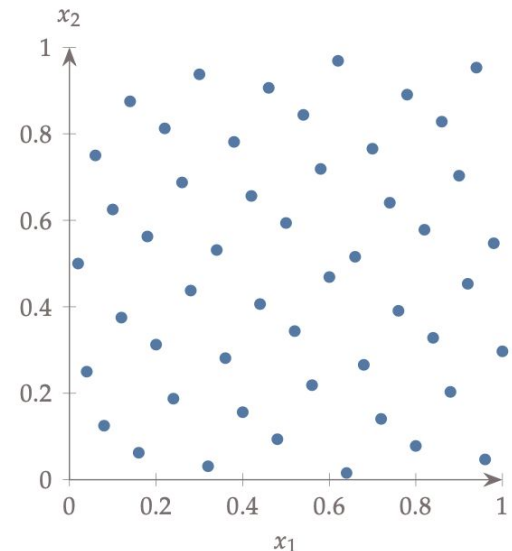
- Full factorial sampling - Not efficient because it scales exponentially with the number of input variables.
- Latin hypercube sampling
- Low-Discrepancy Sequences
 - Halton Sequence
 - Hammersley Sequence



Random



Latin hypercube sampling

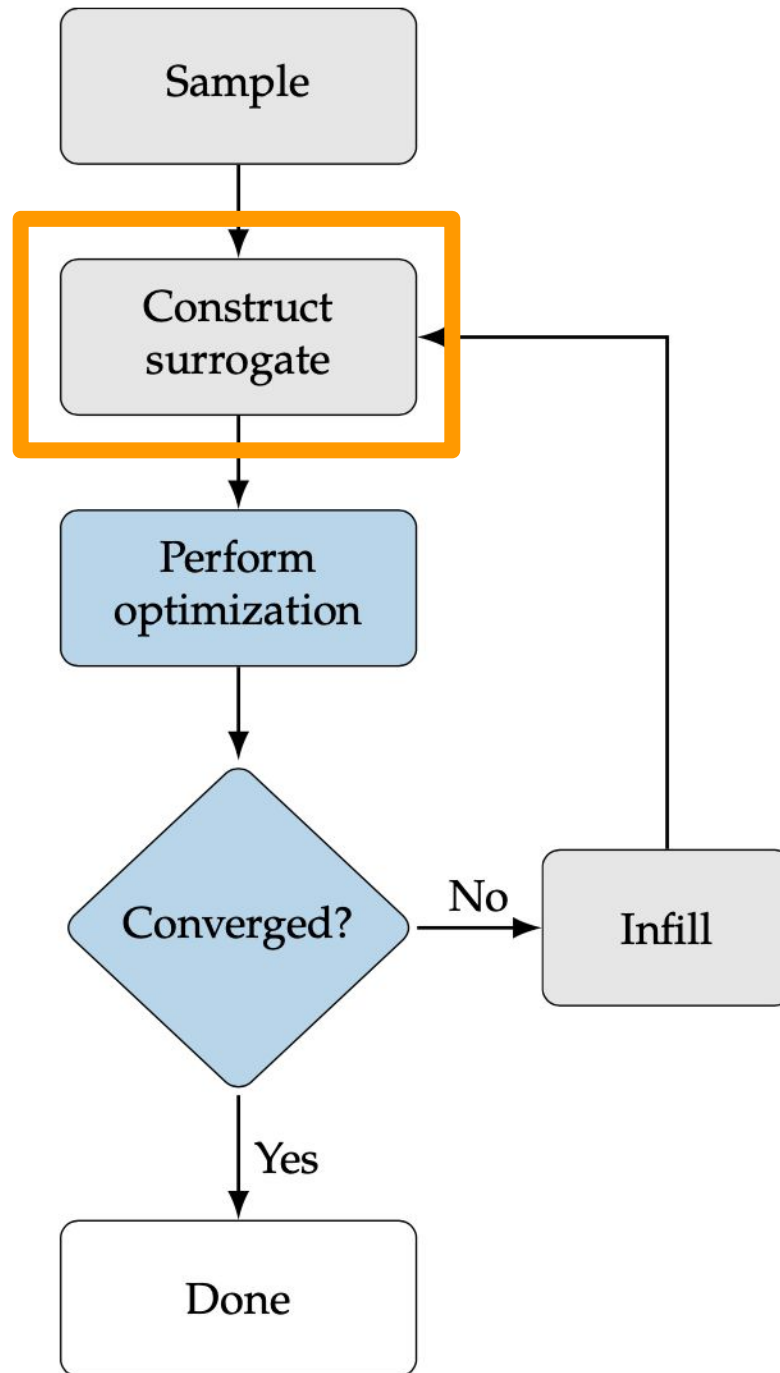


Hammersley sequence with base 2
for the x_2 -axis.

Sampling

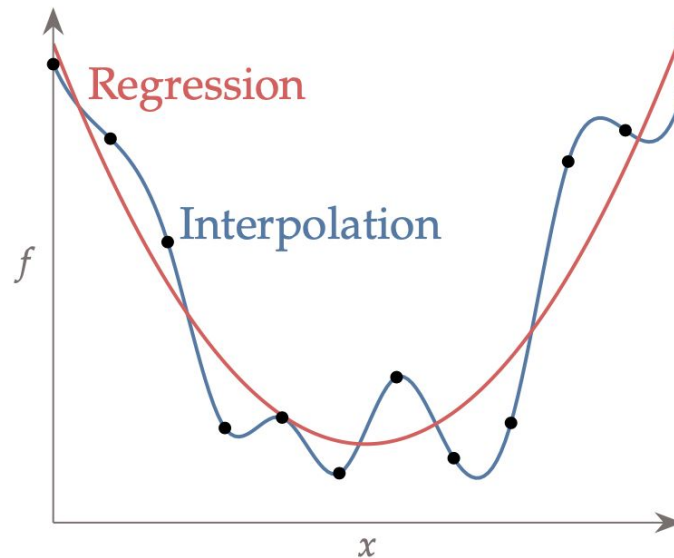
- Sampling strategies on the previous slide are useful in many other applications:
 - Initializing a genetic algorithm
 - Particle swarm optimization
 - Multistart gradient-based algorithm
 - Choosing the points to run in a Monte Carlo simulation
- Because the function behaviour at each sample is independent, we can efficiently parallelise the evaluation of the functions.

Surrogate Process



Constructing Surrogates

- **Regression** models do not try to match the training data points exactly; instead, they minimize the error between a smooth trend function and the training data. Useful when the data are noisy.
- **Interpolation** builds a function that exactly matches the provided training data. Useful when the data are highly multimodal (and not noisy). This is because regression may smooth over variations that are actually physical.



Linear Least Squares Regression

- A linear regression model does not mean that the surrogate is linear in the input variables but rather that the model is **linear in its coefficients** (i.e., linear in the parameters we are estimating). For example:

$$\hat{f}(x) = w_1 x_1^2 + w_2 x_1 x_2 + w_3 \exp(x_2) + w_4 x_1 + w_5$$

- In general

$$\hat{f} = w^\top \psi(x) = \sum_i w_i \psi_i(x)$$

where w is a vector of weights, and ψ is a vector of **basis functions**.

Basis 1 - Polynomial

- Consider a quadratic fit: $f(x) = ax^2 + bx + c$. This can be posed as a linear regression model where the coefficients we wish to estimate are $w = [a, b, c]$ and the basis functions are $\psi = [x^2, x, 1]$.
- For a n -dimensional polynomial model, the basis functions would be polynomials with terms combining the dependencies on all input variables x up to the n -th order. For example, for two input variables up to second order, the basis functions would be

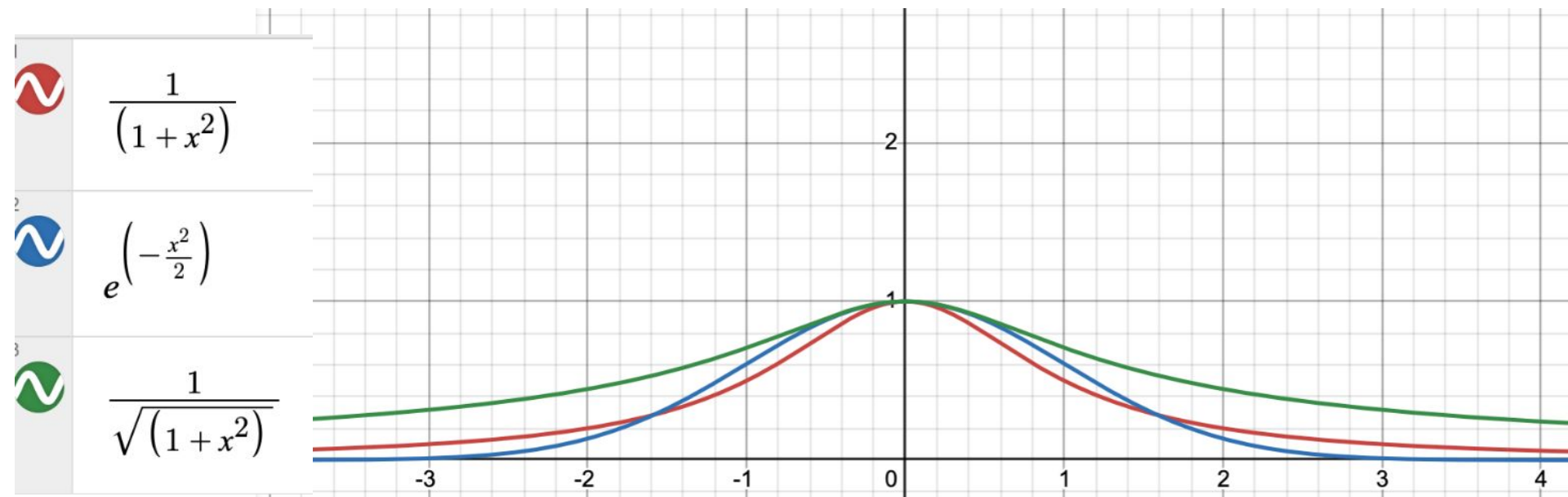
and w would consist of six coefficients.

- In general, the basis functions can be any set of functions. It is usually desirable for these functions to be **orthogonal**.

Basis 2 - Radial Basis Functions (RBF)

- RBFs are functions whose values depend only on the distance between the input and some fixed point.
- Captures the idea that ability to predict function behavior is related to how close we are to known function values.

$$\psi^{(i)} = \psi \left(\|x - c^{(i)}\| \right) = \psi \left(r^{(i)} \right)$$



Basis 2 - Radial Basis Functions (RBF)

Reconstruction and Representation of 3D Objects with Radial Basis Functions

J. C. Carr^{1,2}

R. K. Beatson²

J. B. Cherrie¹

T. J. Mitchell^{1,2}

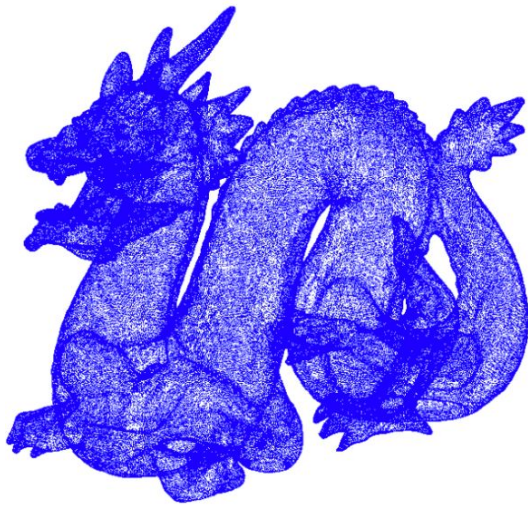
W. R. Fright¹

B. C. McCallum¹

T. R. Evans¹

¹Applied Research Associates NZ Ltd *

²University of Canterbury[†]



(a)



(b)



Figure 1: (a) Fitting a Radial Basis Function (RBF) to a 438,000 point-cloud. (b) Automatic mesh repair using the biharmonic RBF.

<https://www.cs.jhu.edu/~misha/Fall05/Papers/carr01.pdf>

Basis 2 - Radial Basis Functions (RBF)



ARANZ (Applied Research Associates NZ Ltd) was established in 1995 as a private-sector research & development organization, in Christchurch New Zealand.

FastScan, an innovative handheld 3-D laser scanner, and FastRBF, groundbreaking software to model very large 3-D datasets, were the original developments that led ARANZ into many and diverse markets, from geological modelling and mining, to medical imaging and the movie industry.

ARANZ has now morphed into two new and separate companies:

- ARANZ Geo Ltd, that develops and markets Leapfrog suite of products for the mining, hydrogeology, and geothermal industries
- ARANZ Medical Ltd, that develops and markets Silhouette wound imaging, measurement and documentation systems.

All enquiries about ARANZ technologies and market activities should be directed to one or both of these organisations:

- www.aranzgeo.com
- www.aranzmedical.com

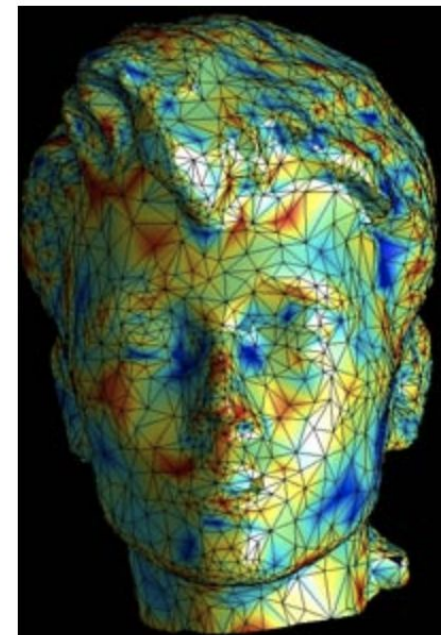
<http://www.aranz.co.nz/>

Initiatives

Silhouette

www.aranzmedical.com

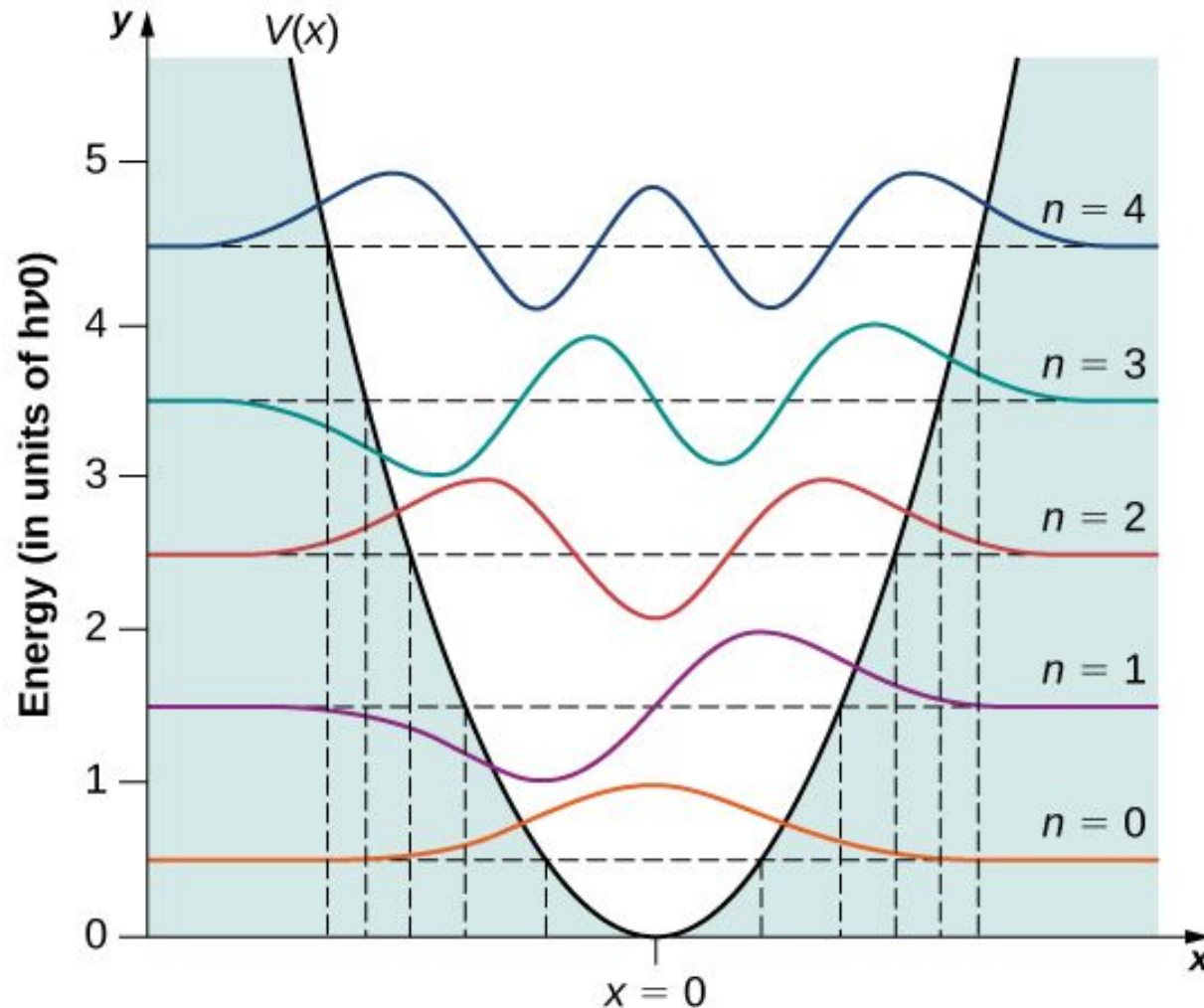
Leapfrog



| aranz technology

Thank you.

Aside: Quantum Harmonic Potential



First four harmonic oscillator normalized wavefunctions

$$\Psi_0 = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-y^2/2}$$

$$\Psi_1 = \left(\frac{\alpha}{\pi}\right)^{1/4} \sqrt{2}y e^{-y^2/2}$$

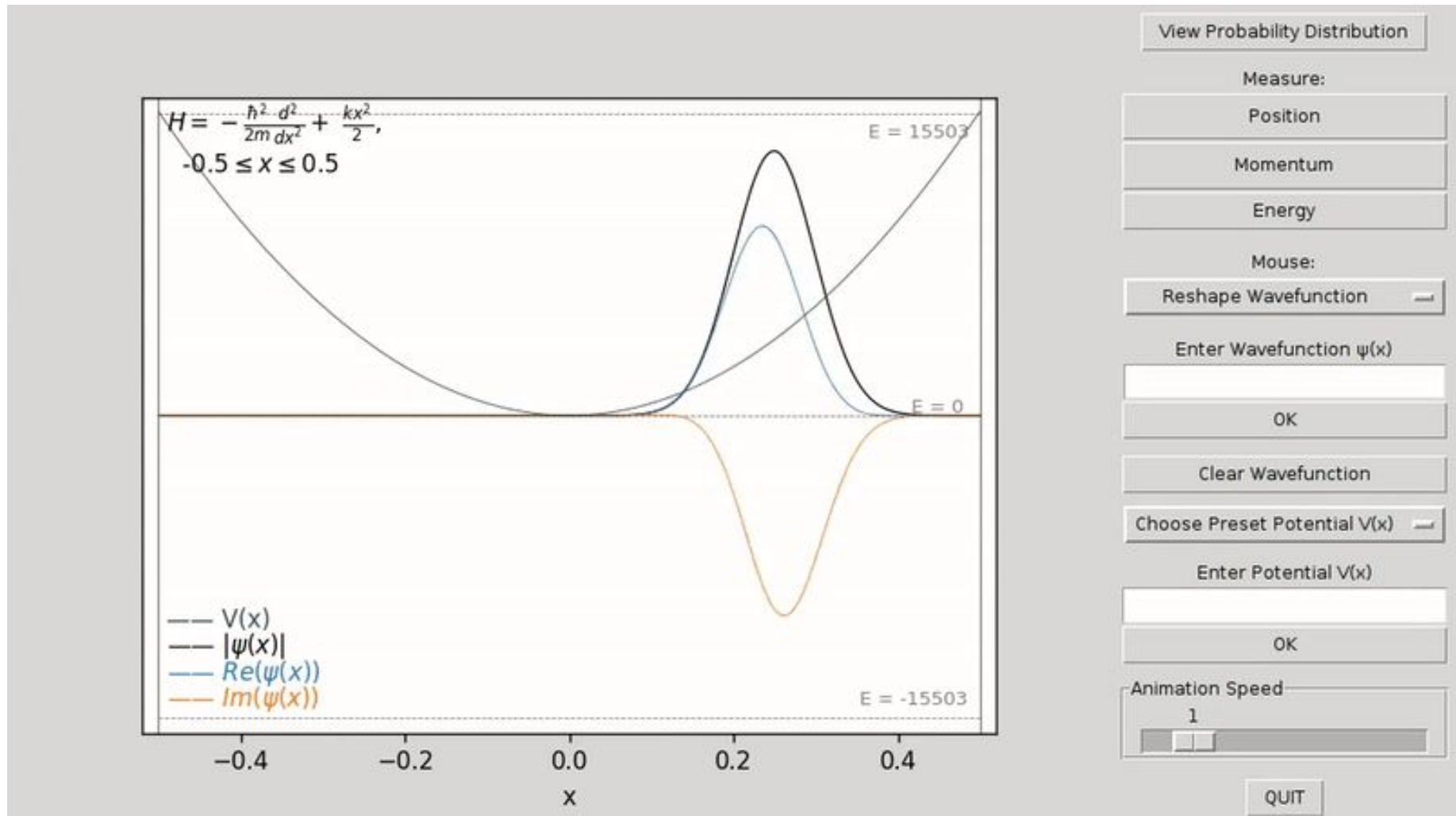
$$\Psi_2 = \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{\sqrt{2}}(2y^2 - 1)e^{-y^2/2}$$

$$\Psi_3 = \left(\frac{\alpha}{\pi}\right)^{1/4} \frac{1}{\sqrt{3}}(2y^3 - 3y)e^{-y^2/2}$$

$$\alpha = \frac{m\omega}{\hbar} \quad y = \sqrt{\alpha} x$$

Aside: Quantum Harmonic Potential

- Wavepacket evolving in time in the harmonic well



Least Squares Solution

- We want to choose coefficients to minimise the error between our predicted function values $\hat{f}$ and the actual function values $f^{(i)}$

$$\underset{w}{\text{minimize}} \quad \sum_i \left(\hat{f}(w; x^{(i)}) - f^{(i)} \right)^2$$

- The solution to this optimisation problem is called a **least squares solution**.

Linear Case

- If the regression model is linear, we can simplify this objective and write the optimisation in matrix form as

$$\underset{w}{\text{minimize}} \quad \|\Psi w - f\|_2^2$$

- Example:

$$\hat{f}(x) = w_1 x_1^2 + w_2 x_1 x_2 + w_3 \exp(x_2) + w_4 x_1 + w_5$$

Linear Case

- The optimisation on the previous slide becomes a quadratic programming problem, with the solution:

$$w = (\Psi^T \Psi)^{-1} \Psi^T f$$

Nonlinear Case

- A **nonlinear least squares regression** model means that the model is nonlinear in the parameters we are estimating.
Example:

$$\hat{f}(x) = w_1 x_1^{w_6} + w_2 x_1 x_2 + w_3 \exp(w_7 x_2) + w_4 x_1 + w_5$$

- We still seek to minimise the sum of errors

$$\underset{w}{\text{minimize}} \quad \sum_i \left(\hat{f}(w; x^{(i)}) - f^{(i)} \right)^2$$

where w is a vector of parameters.

- Use methods like **Levenberg-Marquardt**

Regularisation

- A common variation is to use **regularised least squares** which adds a term in the objective:

$$\underset{w}{\text{minimize}} \quad \|\Psi w - f\|_2^2 + \mu \|w\|_2^2$$

- The rationale is that we may want to accept a higher error in exchange for smaller values of the coefficients.
- A related extension uses a second term of the form $\|w - w_0\|_2$. The idea is that we want a good fit, while maintaining parameters that are close to some known nominal values w_0 .
- This approach is particularly beneficial if the data contain strong *outliers* or are particularly *noisy*.

Regularisation

- Can rewrite the above as

$$\underset{w}{\text{minimize}} \quad \left\| \begin{bmatrix} \Psi \\ \sqrt{\mu}I \end{bmatrix} w - \begin{bmatrix} f \\ 0 \end{bmatrix} \right\|_2^2$$

- This is of the same linear form as before ($\|Aw - b\|^2$), so we can write

$$\begin{aligned} w^* &= (A^\top A)^{-1} A^\top b \\ &= (\Psi^\top \Psi + \mu I)^{-1} \Psi^\top f. \end{aligned}$$

- Also known as:
 - **Ridge regression**
 - **Tikhonov regularization**

Outline

- Multi-Objective Optimisation
- Pareto Optimality and Finding the Pareto Front
- Surrogate-Based Optimisation
- **Maximum Likelihood**

Maximum Likelihood Interpretation

- It can be shown that $\min \text{sum squared error} = \max \text{likelihood}$ under certain conditions

Exercises

- Look up all the terms and derivations that are new to you in this lecture and understand them.
- Think of an optimisation situation when surrogates would be useful.
- Think of another way to commute the Pareto front.

Thanks!